

Additional file 2. Pathway definitions

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MSigDB aggregate components

- Immune: HALLMARK_INTERFERON_GAMMA_RESPONSE, HALLMARK_INTERFERON_ALPHA_RESPONSE, HALLMARK_INFLAMMATORY_RESPONSE, HALLMARK_ALLOGRAFT_REJECTION, HALLMARK_COMPLEMENT
- Proliferation: HALLMARK_E2F_TARGETS, HALLMARK_G2M_CHECKPOINT, HALLMARK_MYC_TARGETS_V1, HALLMARK_MYC_TARGETS_V2, HALLMARK_MITOTIC_SPINDLE
- DNA_Repair: HALLMARK_DNA_REPAIR, REACTOME_HDR_THROUGH_HOMOLOGOUS_RECOMBINATION_HRR, REACTOME_NUCLEOTIDE_EXCISION_REPAIR, REACTOME_MISMATCH_REPAIR, KEGG_MEDICUS_REFERENCE_HOMOLOGOUS_RECOMBINATION, KEGG_MEDICUS_REFERENCE_MISMATCH_REPAIR
- Metabolism: HALLMARK_OXIDATIVE_PHOSPHORYLATION, HALLMARK_GLYCOLYSIS, HALLMARK_FATTY_ACID_METABOLISM, KEGG_MEDICUS_REFERENCE_GLYCOLYSIS
- Stromal_EMT: HALLMARK_EPITHELIAL_MESENCHYMAL_TRANSITION, HALLMARK_ANGIOGENESIS, HALLMARK_HEDGEHOG_SIGNALING
- Apoptosis_Stress: HALLMARK_APOPTOSIS, HALLMARK_P53_PATHWAY, HALLMARK_HYPOXIA
- Hormone: HALLMARK_ESTROGEN_RESPONSE_EARLY, HALLMARK_ESTROGEN_RESPONSE_LATE, HALLMARK_ANDROGEN_RESPONSE

Legacy curated 45-gene panel retained for transparency

- Proliferation: MKI67, AURKA, BIRC5, CCNB1, MYBL2, PLK1, BUB1, CDC20, MCM2, PCNA
- Estrogen_Response: ESR1, PGR, FOXA1, GATA3, XBP1, CA12
- Immune_Response: CD8A, CD4, PDCD1, CD274, CTLA4, FOXP3, IL2RA
- Invasion_EMT: VIM, CDH2, FN1, MMP9, TWIST1, SNAI1, ZEB1
- Apoptosis: CASP3, CASP8, BAX, BAK1, BAD, TP53
- HER2_Signaling: ERBB2, GRB7, PGAP3, MIEN1, STARD3
- Angiogenesis: VEGFA, KDR, FLT1, FGF2